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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/704,167	04/24/2024	Hans KRAEMER	70028US01	5544

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EXAMINER	
POON, DANA LEE	

ART UNIT	PAPER NUMBER
3723	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/704,167	Applicant(s) KRAEMER, Hans	
	Examiner DANA L POON	Art Unit 3723	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 14 July, 2026.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 15-35 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 15-35 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 24 April 2024 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☐ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claim Objections

2. **Claim 27** is objected to because of the following informalities:

- Claim 27 recites "a material having an elongation at break of 500 to 800" that should be "a material having an elongation at break of 500 to 800%".

3. Applicant is advised that should claims 18, 28, and 32 be found allowable, claims 31, 35 and 34, respectively will be objected to under 37 CFR 1.75 as being a substantial duplicate thereof. When two claims in an application are duplicates or else are so close in content that they both cover the same thing, despite a slight difference in wording, it is proper after allowing one claim to object to the other as being a substantial duplicate of the allowed claim. See MPEP § 608.01(m).

Claim Rejections - 35 USC § 103

4. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

5. **Claims 15-23, 25-26 and 28-35** are rejected under 35 U.S.C. 103 as being unpatentable over Kraemer (2009/0313779).

- Regarding Claim 15, Kraemer teaches a toothbrush (Fig. 1) with a handle (Ref. 4, Fig. 1), at least one cleaning zone (Ref. 4, Fig. 1) and at least one deformable element (Ref. 6,

Fig. 4), where the at least one deformable element has a first flexible arm (Ref. 8, Fig. 2-3) arranged towards the at least one cleaning zone (4) and a second flexible arm (Ref. 9, Fig. 4-3) arranged away from the at least one cleaning zone (4, Fig. 4), where the at least one deformable element can move from a resting position (Fig. 4) in which the first flexible arm (8) is flat and aligned to the handle (Fig. 4), to a cleaning position (Fig. 3) in which the first flexible arm (8) and the at least one cleaning zone (4) adopt a curvature of a surface of a dentition (Ref. 20, Fig. 3), where the first flexible arm and second flexible arm form a wedge (Fig .4) being connected together at one end of the at least one deformable element (Fig. 4 show the arms (8&9) are connected on one end (right side))and distanced from each other at the other end of the at least one deformable element (Fig. 4 show the arms (8&9) are connected on one end (left side)), where the first flexible arm and second flexible arm are linked by at least two bridges (Ref. 10, Fig. 4, [0086]) which are tiltable with respect to the first flexible arm and second flexible arm when pressure is applied to the at least one cleaning zone (fig. 3, [0097]), wherein the first flexible arm and second flexible arm and at least two bridges enclose a void space (Ref. 19, Fig. 1-2 show void spaces between the bridges (22), [0098]).

Kraemer teaches wherein the at least one deformable element (6) is not longer than the length of the at least one cleaning zone but rather is the same length as the cleaning zone (Fig. 2A&4, [0098] describes “The deformable element 6 extends essentially only in the cleaning area 4 of the toothbrush 1,…” the length in a longitudinal direction, is not longer than 100% of the cleaning zone). Given Kraemer further teaches the deformable element (6) can be smaller or larger ([0098]), it would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to modify the length of the at least one cleaning zone of Kraemer to

not be longer than 150% of the length of the at least one cleaning zone since it has been held that "[i]n the case where the claimed ranges 'overlap or lie inside ranges disclosed by the prior art' a prima facie case of obviousness exists." *In re Wertheim*, 541 F.2d 257, 191 USPQ 90 (CCPA 1976); *In re Woodruff*, 919 F.2d 1575, 16 USPQ2d 1934 (Fed. Cir. 1990). Further, applicant appears to have placed no criticality on the claimed range (see [Pg. 3, Lines 18-19] that merely states the length is not longer than 150% of the length). Further, such a modification would provide the benefit of enabling novel deformation modes the flexibility characteristics of the toothbrush of the invention may also improve access to the teeth in other way ([0007]).

- Regarding Claim 16, Kraemer teaches the limitations of claim 15, as described above, and further teaches wherein the first flexible arm and second flexible arm and the at least two bridges are formed integrally by injection moulding ([0044&0111] describes elastic elements can be made by injection molding).

- Regarding Claim 17, Kraemer teaches the limitations of claim 15, as described above, and further teaches wherein the first flexible arm and second flexible arm and the at least two bridges are made of a thermoplastic elastomer (TPE) material ([0025,0027,&0081] describes the arms (8&10) and bridges (8) are made of plastic materials and suitable plastic materials are a thermoplastic elastomer).

- Regarding Claim 18, Kraemer teaches the limitations of claim 15, as described above, and further teaches wherein the TPE material has a Shore D hardness ([0027]). Kraemer fails to explicitly teach a Shore D hardness of about 50 to 66. Kraemer discloses that the shore D hardness amount allows the bridges to be bent (Fig. 3) and needs to be optimized to "conform to users teeth." As seen in Fig. 3, the bridges bend and move to conform to users teeth and the

device is disclosed to be a result effective variable in that changing shore D hardness affects guidance and ease of conformity to teeth. Further, it appears that one of ordinary skill in the art would have had a reasonable expectation of success in modifying the Kraemer reference to have a shore D hardness within the claimed range, as it involves only adjusting the material property of a component disclosed. Therefore, it would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to modify the device of Kraemer by making the shore D hardness 50 to 66 as a matter of routine optimization since it has been held that "where the general conditions of a claim are disclosed in the prior art, it is not inventive to discover the optimum or workable ranges by routine experimentation." *In re Aller*, 220 F.2d 454, 456, 105 USPQ 233, 235 (CCPA 1955). Further changing Kraemer's shore D hardness produces the same predictable result of allowing the bridges to bend to conform to teeth as required in the instant application. Further, applicant appears to have placed no criticality on the claimed range ([Pg. 6, Lines 20-22] merely states the TPE material to be a plethora of shore D hardnesses).

- Regarding Claim 19, Kraemer teaches the limitations of claim 15, as described above, and further teaches further teaches wherein the bridges have hinges (Ref. 15&17, Fig. 2A) flexibly connecting the bridges to the arms (Fig. 2A-3, [0087]).

- Regarding Claim 20, Kraemer teaches the limitations of claim 15, as described above, and further teaches wherein the hinges are integral hinges (Fig. 2A).

- Regarding Claim 21, Kraemer teaches the limitations of claim 15, as described above, and further teaches wherein the integral hinges have a thickness in the longitudinal direction of the toothbrush (Fig. 2A, [0087] describes the hinges to be thinner than the bridges to allow for deformation). Kraemer fails to explicitly teach the thickness of the hinges to be 0.2 to 0.4mm.

Kraemer discloses that the thickness of the hinges allows the bridges to be bent (Figs. 2A&3,[0087]) and needs to be optimized to "conform to users teeth." As seen in Fig. 2A&3, the thickness of the hinges allows the bridges to bend to conform to users teeth and the device is disclosed to be a result effective variable in that the thickness affects guidance and ease of conformity to teeth. Further, it appears that one of ordinary skill in the art would have had a reasonable expectation of success in modifying the Kraemer reference to have a thickness within the claimed range, as it involves only adjusting the dimension of a component disclosed to require adjustment. Therefore, it would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to modify the device of Kraemer by making the thickness of the hinges to be 0.2mm to 0.4mm as a matter of routine optimization since it has been held that "where the general conditions of a claim are disclosed in the prior art, it is not inventive to discover the optimum or workable ranges by routine experimentation." *In re Aller*, 220 F.2d 454, 456, 105 USPQ 233, 235 (CCPA 1955). Further changing the thickness of the hinges in Kraemer produces the same predictable result of allowing the bridges to bend to conform to teeth as required in the instant application.

- Regarding Claim 22, Kraemer teaches the limitations of claim 15, as described above, and further teaches wherein a distance between the arms (Fig. 4 shows distance between the arms (8&9)) at a proximal end of the deformable element (Fig. 4 left and right sides without hashing). Kraemer fails to explicitly teach the distance to be between 3mm to 5mm. Kraemer discloses that the distance between the arms amount allows the bridges to be placed between and bent (Fig. 3) and needs to be optimized to "conform to users teeth." As seen in Fig. 3, the distance between the arms to conform to users teeth and the device is disclosed to be a result effective variable in that the distance affects guidance and ease of conformity to teeth. Further, it appears that one of ordinary skill in the art would have had a reasonable expectation of success

in modifying the Kraemer reference to have a distance within the claimed range, as it involves only adjusting the dimension of a component disclosed to require adjustment. Therefore, it would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to modify the device of Kraemer by making the distance of between the arms to be 3mm to 5mm as a matter of routine optimization since it has been held that "where the general conditions of a claim are disclosed in the prior art, it is not inventive to discover the optimum or workable ranges by routine experimentation." *In re Aller*, 220 F.2d 454, 456, 105 USPQ 233, 235 (CCPA 1955). Further changing the distance in Kraemer produces the same predictable result of allowing the bridges to bend to conform to teeth as required in the instant application.

- Regarding Claim 23, Kraemer teaches the limitations of claim 15, as described above, and further teaches wherein there is a longitudinal distance between adjacent bridges (Fig. 4 shows distance between the bridges). Kraemer fails to explicitly teach the distance to be between 5mm to 9mm. Kraemer discloses that the distance between the bridges allows the deformable region to be bent (Fig. 3) and needs to be optimized to "conform to users teeth." As seen in Fig. 3, the distance between the bridges to conform to users teeth and the device is disclosed to be a result effective variable in that the distance affects guidance and ease of conformity to teeth. Further, it appears that one of ordinary skill in the art would have had a reasonable expectation of success in modifying the Kraemer reference to have a distance within the claimed range, as it involves only adjusting the dimension of a component disclosed to require adjustment. Therefore, it would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to modify the device of Kraemer by making the distance of between adjacent bridges to be 5mm to 9mm as a matter of routine optimization since it has been held that "where the general conditions of a claim are disclosed in the prior art, it is not

inventive to discover the optimum or workable ranges by routine experimentation." *In re Aller*, 220 F.2d 454, 456, 105 USPQ 233, 235 (CCPA 1955). Further changing the distance in Kraemer produces the same predictable result of allowing the bridges to bend to conform to teeth as required in the instant application.

- Regarding Claim 25, Kraemer teaches the limitations of claim 15, as described above, and further teaches wherein the at least one deformable element has 2-7 bridges, preferably 3-5 bridges, most preferably 3 bridges (Fig. 2).

- Regarding Claim 26, Kraemer teaches the limitations of claim 15, as described above, and further teaches wherein the handle comprises a polypropylene skeleton and is formed integrally in one piece with the at least one deformable element by injection moulding ([0050-0052 & 0081-0082] describes the handle and wing are integrally made of polypropylene by injection molding forming a plastic skeleton).

- Regarding Claim 28, Kraemer teaches the limitations of claim 15, as described above, and further teaches wherein an angle (a) between the at least two bridges and the first flexible arm (8) and second flexible arm in the resting position (Fig. 2A&4 show the bridges are about perpendicular), and an angle obtuse when pressure is applied to the at least one cleaning zone (Fig. 3 shows an obtuse angle). However, Kraemer fails to specifically teach an angle between the at least two bridges and the first flexible arm and second flexible arm is 90 degrees in the resting position, and 95 to 110 when pressure is applied to the at least one cleaning zone. It would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to modify the angle between the bridges and flexible arms of Kraemer to not be 90 degrees in the resting position, and 95 to 110 degrees when pressure is applied to the

at least one cleaning zone since it has been held that "[i]n the case where the claimed ranges 'overlap or lie inside ranges disclosed by the prior art' a prima facie case of obviousness exists." *In re Wertheim*, 541 F.2d 257, 191 USPQ 90 (CCPA 1976); *In re Woodruff*, 919 F.2d 1575, 16 USPQ2d 1934 (Fed. Cir. 1990). Further, applicant appears to have placed no criticality on the claimed range (see [Pg. 6, Lines 13-16] that merely states the angle varies from 90 degrees when no pressure is applied up to 110 degrees to during cleaning when pressure is applied).

- Regarding Claim 29, Kraemer teaches toothbrush (Fig. 1) with a handle (Ref. 4, Fig. 1), at least one cleaning zone (Ref. 4, Fig. 1) and at least one deformable element (Ref. 6, Fig. 4), where the at least one deformable element has a first flexible arm (Ref. 8, Fig. 2-3) arranged towards the at least one cleaning zone (4) and a second flexible arm (Ref. 9, Fig. 4-3) arranged away from the at least one cleaning zone (4, Fig. 4), where the at least one deformable element can move from a resting position (Fig. 4) in which the first flexible arm (8) is flat and aligned to the handle (Fig. 4), to a cleaning position (Fig. 3), to a cleaning position (Fig. 3) in which the first flexible arm (8) and the at least one cleaning zone adopt a curvature of a surface of a dentition (Ref. 20, Fig. 3), where the first flexible arm and second flexible arm form a wedge (Fig .4) being connected together at one end of the at least one deformable element (Fig. 4 show the arms (8&9) are connected on one end (right side)) and distanced from each other at the other end of the at least one deformable element (Fig. 4 show the arms (8&9) are connected on one end (left side)), where the first flexible arm and second flexible arm are linked by at least two bridges (Ref. 10, Fig. 4, [0086]) which are tiltable with respect to the first flexible arm and second flexible arm when pressure is applied to the at least one cleaning zone (fig. 3, [0097]), wherein the first flexible arm and second flexible arm and at least two bridges enclose a void space (Ref. 19, Fig. 1-2 show void spaces between the bridges (22), [0098]).

Kraemer further teaches wherein an angle (a) between the at least two bridges and the first flexible arm (8) and second flexible arm in the resting position (Fig. 2A&4 show the bridges are about perpendicular, [0097] describes " the guide elements 10 can be arranged in the resting position at an angle of 90 degrees to the axis L-L"), and an angle obtuse when pressure is applied to the at least one cleaning zone (Fig. 3 shows an obtuse angle, [0094] describes "...when a cleaning force F is being exerted on the cleaning zone 4 of the toothbrush...a comparison between the cleaning position and the resting position, the contour of the deformable element 6 from FIG. 2 is shown in FIG. 3 as a dotted line", the angle is changed when a cleaning force is applied to the cleaning zone). Given Kraemer further teaches the bridge elements can be a range of angles ([0024]), it would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to modify the angle between the bridges and flexible arms of Kraemer to not be 90 degrees in the resting position, and 95 to 110 degrees when pressure is applied to the at least one cleaning zone since it has been held that "[i]n the case where the claimed ranges 'overlap or lie inside ranges disclosed by the prior art' a prima facie case of obviousness exists." *In re Wertheim*, 541 F.2d 257, 191 USPQ 90 (CCPA 1976); *In re Woodruff*, 919 F.2d 1575, 16 USPQ2d 1934 (Fed. Cir. 1990). Further, applicant appears to have placed no criticality on the claimed range (see [Pg. 6, Lines 13-16] that merely states the angle varies from 90 degrees when no pressure is applied up to 110 degrees to during cleaning when pressure is applied). Further, the deformation allows more effective brushing by allowing to reach premolars easier ([0097]).

- Regarding Claim 30, Kraemer teaches the limitations of claim 29, as described above, and further teaches wherein the first flexible arm and second flexible arm and the at least two bridges are made of a thermoplastic elastomer (TPE) material ([0025,0027,&0081] describes the

arms (8&10) and bridges (8) are made of plastic materials and suitable plastic materials are a thermoplastic elastomer).

- Regarding Claim 31, Kraemer teaches the limitations of claim 30, as described above, and further teaches wherein the TPE material has a Shore D hardness ([0027]). Kraemer fails to explicitly teach a Shore D hardness of about 50 to 66. Kraemer discloses that the shore D hardness amount allows the bridges to be bent (Fig. 3) and needs to be optimized to "conform to users teeth." As seen in Fig. 3, the bridges bend and move to conform to users teeth and the device is disclosed to be a result effective variable in that changing shore D hardness affects guidance and ease of conformity to teeth. Further, it appears that one of ordinary skill in the art would have had a reasonable expectation of success in modifying the Kraemer reference to have a shore D hardness within the claimed range, as it involves only adjusting the material property of a component disclosed. Therefore, it would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to modify the device of Kraemer by making the shore D hardness 50 to 66 as a matter of routine optimization since it has been held that "where the general conditions of a claim are disclosed in the prior art, it is not inventive to discover the optimum or workable ranges by routine experimentation." *In re Aller*, 220 F.2d 454, 456, 105 USPQ 233, 235 (CCPA 1955). Further changing Kraemer's shore D hardness produces the same predictable result of allowing the bridges to bend to conform to teeth as required in the instant application. Further, applicant appears to have placed no criticality on the claimed range ([Pg. 6, Lines 20-22] merely states the TPE material to be a plethora of shore D hardnesses).

- Regarding Claim 32, Kraemer teaches the limitations of claim 29, as described above, and Kraemer teaches wherein the at least one deformable element (6) is not longer

than the length of the at least one cleaning zone but rather is the same length as the cleaning zone (Fig. 2A&4, [0098] describes "The deformable element 6 extends essentially only in the cleaning area 4 of the toothbrush 1,..." the length in a longitudinal direction, is not longer than 100% of the cleaning zone). Given Kraemer further teaches the deformable element (6) can be smaller or larger ([0098]), it would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to modify the length of the at least one cleaning zone of Kraemer to not be longer than 150% of the length of the at least one cleaning zone since it has been held that "[i]n the case where the claimed ranges 'overlap or lie inside ranges disclosed by the prior art' a prima facie case of obviousness exists." *In re Wertheim*, 541 F.2d 257, 191 USPQ 90 (CCPA 1976); *In re Woodruff*, 919 F.2d 1575, 16 USPQ2d 1934 (Fed. Cir. 1990). Further, applicant appears to have placed no criticality on the claimed range (see [Pg. 3, Lines 18-19] that merely states the length is not longer than 150% of the length). Further, such a modification would provide the benefit of enabling novel deformation modes the flexibility characteristics of the toothbrush of the invention may also improve access to the teeth in other way ([0007]).

- Regarding Claim 33, Kraemer teaches A toothbrush (Fig. 1) with a handle (Ref. 4, Fig. 1), at least one cleaning zone (Ref. 4, Fig. 1) and at least one deformable element (Ref. 6, Fig. 4), where the at least one deformable element has a first flexible arm (Ref. 8, Fig. 2-3) arranged towards the at least one cleaning zone (4) and a second flexible arm (Ref. 9, Fig. 4-3) arranged away from the at least one cleaning zone (4, Fig. 4), where the at least one deformable element can move from a resting position (Fig. 4) in which the first flexible arm (8) is flat and aligned to the handle (Fig. 4), to a cleaning position (Fig. 3), to a cleaning position (Fig. 3) in which the first flexible arm (8) and the at least one cleaning zone adopt a curvature of a surface of a dentition (Ref. 20, Fig. 3),

where the first flexible arm and second flexible arm form a wedge (Fig .4) being connected together at one end of the at least one deformable element (Fig. 4 show the arms (8&9) are connected on one end (right side)) and distanced from each other at the other end of the at least one deformable element (Fig. 4 show the arms (8&9) are connected on one end (left side)), where the first flexible arm and second flexible arm are linked by at least two bridges (Ref. 10, Fig. 4, [0086]) which are tiltable with respect to the first flexible arm and second flexible arm when pressure is applied to the at least one cleaning zone (fig. 3, [0097]), wherein the first flexible arm and second flexible arm and at least two bridges enclose a void space (Ref. 19, Fig. 1-2 show void spaces between the bridges (22), [0098]), wherein the first flexible arm and second flexible arm are made of a thermoplastic elastomer (TPE) material ([0025,0027,&0081] describes the arms (8&10) and bridges (8) are made of plastic materials and suitable plastic materials are a thermoplastic elastomer). Kraemer further teaches further teaches wherein the TPE material has a Shore D hardness ([0025&0027]). Kraemer fails to explicitly teach a Shore D hardness of about 50 to 66. Kraemer discloses that the shore D hardness amount allows the bridges to be bent (Fig. 3) and needs to be optimized to "conform to users teeth." As seen in Fig. 3, the bridges bend and move to conform to users teeth and the device is disclosed to be a result effective variable in that changing shore D hardness affects guidance and ease of conformity to teeth. Further, it appears that one of ordinary skill in the art would have had a reasonable expectation of success in modifying the Kraemer reference to have a shore D hardness within the claimed range, as it involves only adjusting the material property of a component disclosed. Therefore, it would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to modify the device of Kraemer by making the shore D hardness 50 to 66 as a matter of routine optimization since it has been held that "where the general conditions of a claim are disclosed in the prior art, it is not inventive to

discover the optimum or workable ranges by routine experimentation." *In re Aller*, 220 F.2d 454, 456, 105 USPQ 233, 235 (CCPA 1955). Further changing Kraemer's shore D hardness produces the same predictable result of allowing the bridges to bend to conform to teeth as required in the instant application. Further, applicant appears to have placed no criticality on the claimed range ([Pg. 6, Lines 20-22] merely states the TPE material to be a plethora of shore D hardnesses).

- Regarding Claim 34, Kraemer teaches the limitations of claim 33, as described above, and Kraemer teaches wherein the at least one deformable element (6) is not longer than the length of the at least one cleaning zone but rather is the same length as the cleaning zone (Fig. 2A&4, [0098] describes "The deformable element 6 extends essentially only in the cleaning area 4 of the toothbrush 1,..." the length in a longitudinal direction, is not longer than 100% of the cleaning zone). Given Kraemer further teaches the deformable element (6) can be smaller or larger ([0098]), it would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to modify the length of the at least one cleaning zone of Kraemer to not be longer than 150% of the length of the at least one cleaning zone since it has been held that "[i]n the case where the claimed ranges 'overlap or lie inside ranges disclosed by the prior art' a prima facie case of obviousness exists." *In re Wertheim*, 541 F.2d 257, 191 USPQ 90 (CCPA 1976); *In re Woodruff*, 919 F.2d 1575, 16 USPQ2d 1934 (Fed. Cir. 1990). Further, applicant appears to have placed no criticality on the claimed range (see [Pg. 3, Lines 18-19] that merely states the length is not longer than 150% of the length). Further, such a modification would provide the benefit of enabling novel deformation modes the flexibility characteristics of the toothbrush of the invention may also improve access to the teeth in other way ([0007]).

- Regarding Claim 35, Kraemer teaches the limitations of claim 33, as described above, and Kraemer further teaches wherein an angle (a) between the at least two bridges and the first flexible arm (8) and second flexible arm in the resting position (Fig. 2A&4 show the bridges are about perpendicular, [0097] describes " the guide elements 10 can be arranged in the resting position at an angle of 90 degrees to the axis L-L"), and an angle obtuse when pressure is applied to the at least one cleaning zone (Fig. 3 shows an obtuse angle, [0094] describes "...when a cleaning force F is being exerted on the cleaning zone 4 of the toothbrush...a comparison between the cleaning position and the resting position, the contour of the deformable element 6 from FIG. 2 is shown in FIG. 3 as a dotted line", the angle is changed when a cleaning force is applied to the cleaning zone). Given Kraemer further teaches the bridge elements can be a range of angles ([0024]), it would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to modify the angle between the bridges and flexible arms of Kraemer to not be 90 degrees in the resting position, and 95 to 110 degrees when pressure is applied to the at least one cleaning zone since it has been held that "[i]n the case where the claimed ranges 'overlap or lie inside ranges disclosed by the prior art' a prima facie case of obviousness exists." *In re Wertheim*, 541 F.2d 257, 191 USPQ 90 (CCPA 1976); *In re Woodruff*, 919 F.2d 1575, 16 USPQ2d 1934 (Fed. Cir. 1990). Further, applicant appears to have placed no criticality on the claimed range (see [Pg. 6, Lines 13-16] that merely states the angle varies from 90 degrees when no pressure is applied up to 110 degrees to during cleaning when pressure is applied). Further, the deformation allows more effective brushing by allowing to reach premolars easier ([0097]).

6. **Claim 27** is rejected under 35 U.S.C. 103 as being unpatentable over Kraemer as applied to claim 15 above, and further in view of Precision Punch and Plastics (NPL, https://precisionpunch.com/plastic-materials-index/polypropylene/).

- Regarding Claim 27, Kraemer teaches the limitations of claim 15, as described above, and further teaches the deformable element is made of polypropylene ([0050-0052&0081]). Kraemer fails to explicitly teach wherein the deformable element consists of a material having an elongation at break of 500 to 800%. Precision and Punch teaches polypropylene having an elongation at break of 100 to 600%. Therefore, It would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to modify the deformable element made of polypropylene having an elongation break from 500 to 800% since it has been held that "[i]n the case where the claimed ranges 'overlap or lie inside ranges disclosed by the prior art' a prima facie case of obviousness exists." *In re Wertheim*, 541 F.2d 257, 191 USPQ 90 (CCPA 1976); *In re Woodruff*, 919 F.2d 1575, 16 USPQ2d 1934 (Fed. Cir. 1990). Further, since the polypropylene is known to have an elongation break that overlaps with the claimed range, it would have been obvious to one of ordinary skill in the art before the effective filing date to modify the elongation at bread to be between 500 to 800% since such a modification would yield the predictable result of allowing the deformable element to conform to teeth.

Response to Arguments

7. Applicant's amendments to the claims are acknowledged and examiner has withdrawn the previous 35 USC 112(b) and claim objections. However, the amendment to the claims yielded a new claim objection.

- Applicant's arguments, filed 14 July, 2026, with respect to the 35 USC 102 rejection of claim 15 have been fully considered and are persuasive. The 35 USC 102 rejection of

claim 15 has been withdrawn. However, Examiner has applied a new 35 USC 103 rejection above. Regarding applicant's arguments that the prior art fails to explicitly teach the at least one deformable element is not longer than 150% of the length of the at least one cleaning zone has been fully considered and is persuasive. Examiner has applied a new 35 USC 103 rejection above. Kraemer teaches wherein the at least one deformable element (6) is not longer than the length of the at least one cleaning zone but rather is the same length as the cleaning zone (Fig. 2A&4, [0098] describes "The deformable element 6 extends essentially only in the cleaning area 4 of the toothbrush 1,..." the length in a longitudinal direction, is not longer than 100% of the cleaning zone). Given Kraemer further teaches the deformable element (6) can be smaller or larger ([0098]), it would have been obvious to one having ordinary skill in the art before the effective filing date of the claimed invention to modify the length of the at least one cleaning zone of Kraemer to not be longer than 150% of the length of the at least one cleaning zone since it has been held that "[i]n the case where the claimed ranges 'overlap or lie inside ranges disclosed by the prior art' a prima facie case of obviousness exists." *In re Wertheim*, 541 F.2d 257, 191 USPQ 90 (CCPA 1976); *In re Woodruff*, 919 F.2d 1575, 16 USPQ2d 1934 (Fed. Cir. 1990). Further, applicant appears to have placed no criticality on the claimed range (see [Pg. 3, Lines 18-19] that merely states the length is not longer than 150% of the length). Further, such a modification would provide the benefit of enabling novel deformation modes the flexibility characteristics of the toothbrush of the invention may also improve access to the teeth in other way ([0007]).

- Regarding claim 27, In response to applicant's argument that there is no teaching, suggestion, or motivation to combine the references, the examiner recognizes that obviousness may be established by combining or modifying the teachings of the prior art to produce the claimed

invention where there is some teaching, suggestion, or motivation to do so found either in the references themselves or in the knowledge generally available to one of ordinary skill in the art. See *In re Fine*, 837 F.2d 1071, 5 USPQ2d 1596 (Fed. Cir. 1988), *In re Jones*, 958 F.2d 347, 21 USPQ2d 1941 (Fed. Cir. 1992), and *KSR International Co. v. Teleflex, Inc.*, 550 U.S. 398, 82 USPQ2d 1385 (2007). In this case, examiner is merely using the NPL to teach that the material polypropylene that the deformable element of Kraemer is made of has an elongation at break of 100 to 600%. The test for obviousness is not whether the features of a secondary reference may be bodily incorporated into the structure of the primary reference; nor is it that the claimed invention must be expressly suggested in any one or all of the references. Rather, the test is what the combined teachings of the references would have suggested to those of ordinary skill in the art. See *In re Keller*, 642 F.2d 413, 208 USPQ 871 (CCPA 1981).

- Regarding Claim 33, applicant's arguments in regards to the prior art fails to teach the Shore D hardness has been fully considered and is not persuasive. Applicant's instant application have placed no criticality on the claimed range or unexpected results for the claimed range (instant application specifications, [Pg. 6, Lines 20-22] merely states the TPE material to be a plethora of shore D hardnesses). While Kraemer does teach having preferable a shore D of 100 or more, it does not teach away or limit the shore D hardness from being different values. Given there is no criticality or unexpected results of the claimed range from the instant application, Kraemer teaches in Fig. 3, the bridges bend and move to conform to users teeth and the device is disclosed to be a result effective variable in that changing shore D hardness affects guidance and ease of conformity to teeth. Further, it appears that one of ordinary skill in the art would have had a reasonable expectation of success in modifying the Kraemer reference to have a shore D hardness within the claimed range, as it involves only adjusting the material property of a component disclosed. Therefore, it would have been obvious to one having ordinary skill in the

art before the effective filing date of the claimed invention to modify the device of Kraemer by making the shore D hardness 50 to 66 as a matter of routine optimization since it has been held that "where the general conditions of a claim are disclosed in the prior art, it is not inventive to discover the optimum or workable ranges by routine experimentation." *In re Aller*, 220 F.2d 454, 456, 105 USPQ 233, 235 (CCPA 1955). Further changing Kraemer's shore D hardness produces the same predictable result of allowing the bridges to bend to conform to teeth as required in the instant application. Further, applicant appears to have placed no criticality on the claimed range ([Pg. 6, Lines 20-22] merely states the TPE material to be a plethora of shore D hardnesses).

Conclusion

8. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure. Kramer (6,327,735), Berl (4,694,844), Blakeman (2,706,825), and Chodorow (WO 0121035 A1) teaches toothbrushes with a deformable element and can be considered analogous art because they are within the same field of endeavor.

9. Any inquiry concerning this communication or earlier communications from the examiner should be directed to DANA L POON whose telephone number is (571)272-6164. The examiner can normally be reached on General: 6:30AM-3:30PM.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice. If attempts to reach the examiner by telephone are unsuccessful, the examiner' s supervisor, David Posigian can be reached on (313) 446-6546. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300. Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for

published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see https://ppairmy.uspto.gov/pair/PrivatePair. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

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